

CropIQ: Enhancing Crop Selection with Satellite Image Analysis

CAPSTONE PROJECT REPORT

Submitted by

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**in partial fulfillment for the award of the degree
of**

BACHELOR OF TECHNOLOGY

IN

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SCHOOL OF COMPUTING

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We affirm that the project work titled “**CropIQ: Enhancing Crop Selection with Satellite image analysis**” being submitted in partial fulfilment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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BONAFIDE CERTIFICATE

Certified that this project report **“CropIQ: Enhancing Crop Selection with Satellite Image Analysis”** is the bonafide work of **“P. Durga Prasadu (9921004520), P. Anjana Sree (9921004816), P. Manideep (9921004564), P. Lohitha Lakshmi (9921004563)”** who carried out the project work under my supervision.

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SCHOOL OF COMPUTING
COMPUTER SCIENCE AND ENGINEERING
PROJECT SUMMARY

Project Title	Crop recommendation system with satellite images using Efficient Net and CNN deep learning	
Project Team Members (Name with Register No)	PADALA. DURGA PRASADU - 9921004520 PEETA. ANJANA SREE – 9921004816 PEYYALA. MANIDEEP – 9921004564 PETLU. LOHITHA LAKSMI - 9921004563	
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Program Concentration Area	Agricultural Informatics and Artificial Intelligence (AI) in Precision Agriculture	
Technical Requirements	Deep Learning, Machine Learning.	
Engineering standards and realistic constraints in these areas		
Area	Codes & Standards / Realistic Constraints	Tick ✓
Economic	Efficient use of computational resources, cost-effective processing, and implementation to remain affordable for small-scale farmers.	✓
Environmental	Compliance with environmental standards to minimize ecological impact and promote sustainable agriculture.	✓
Social	Standards for social responsibility, particularly in supporting rural agricultural communities.	✓
Ethical	Adherence to ethical data usage and transparency standards.	✓
Health and Safety	Standards for health and safety in agricultural recommendations.	x
Manufacturability	Consideration of scalability and easy replication of the model across different regions.	✓
Sustainability	Standards for promoting long-term agricultural and environmental sustainability.	✓

ABSTRACT

The study presents an innovative "soil-based crop recommendation system employing Satellite imagery" that enhances agricultural practices through the utilization of Sentinel-2 images from various regions. By leveraging advanced satellite data analysis, the proposed system discerns crucial details regarding soil types and properties, enabling optimal crop selection based on specific soil conditions. We enhance the model's performance even further by employing the Swish and Mish activation functions. These activations have a reputation for enhancing the learning process within deep neural networks by accelerating convergence and detecting intricate patterns in the data. Utilizing advanced techniques, our model excels in predicting crop recommendations based on soil types with greater accuracy. The main objective of this system is to enhance agricultural productivity and sustainability through customized crop recommendations tailored to specific soil properties. We offer a data-driven solution that offers farmers valuable insights to enhance soil management methods and optimize crop yield in various agricultural areas by combining satellite technology with advanced deep learning techniques. We aim to transform farming methods by empowering stakeholders to make informed decisions that enhance resource efficiency, increase profitability, and foster a sustainable agricultural environment.

Key Words – Efficient Net, Deep-learning, Swish Mish activations, Crop recommendation.

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LIST OF ACADEMIC REFERENCE COURSES

S. NO.	COURSE CODE	COURSE NAME
1.	213CSE3301	DEEP LEARNING
2.	212CSE2304	MACHINE LEARNING
3.	213CSE4301	PATTERN AND ANAMOLY DETECTION
4.	211CSE1402	PYTHON PROGRAMMING
5.	213CSE2301	PREDICTIVE ANALYTICS

CHAPTER – I

INTRODUCTION

1.1 Background and Motivation

Background for CropIQ Project

Agriculture forms the backbone of many economies and sustains the livelihoods of millions globally. Despite its significance, farmers face numerous challenges, including inefficient resource utilization, unpredictable weather, poor soil management, and lack of crop-specific guidance. With the increasing demand for sustainable farming, leveraging technology to enhance agricultural practices has become essential.

Traditional methods for determining crop suitability often rely on manual testing of soil or generic recommendations, which may not consider local soil conditions or climatic factors. This gap between scientific advancements and on-ground practices has created an opportunity for technology-driven solutions to optimize crop selection and boost agricultural productivity.

Recent advancements in satellite imagery, machine learning, and IoT technologies enable precise soil and environmental analysis. CropIQ combines these technologies to bridge the gap between data and actionable insights, providing farmers with accurate crop recommendations based on soil quality, weather conditions, and geographic location.

Motivation for CropIQ Project

The inspiration for CropIQ comes from the need to empower farmers with reliable and real-time data to improve decision-making. The following factors motivated the development of CropIQ:

1. Challenges in Traditional Farming:

- Lack of personalized recommendations for crop selection.
- Reliance on outdated or generic agricultural practices.
- Limited access to expert guidance in rural areas.

2. Technology as an Enabler:

- Advances in machine learning models like EfficientNet and access to satellite imagery provide the potential to analyze soil and recommend crops with high precision.
- Smartphone penetration in rural areas creates an opportunity to deliver these insights through mobile applications.

3. Sustainability and Productivity:

- Encouraging the use of technology to achieve higher crop yields and reduce resource wastage.
- Promoting environmentally friendly practices by recommending crops that suit the soil and climate conditions.

4. Impact on Farmers:

- Helping farmers increase profitability by growing suitable crops.
- Reducing the risk of crop failures by basing recommendations on scientific data.

By integrating cutting-edge technology into agriculture, CropIQ seeks to revolutionize farming practices and contribute to global food security, economic development, and sustainable agriculture.

1.2 Problem Statement

The problem in agriculture today is twofold: traditional crop recommendation methods are insufficient to handle the complexity of modern agriculture, and there is an urgent need for sustainable farming solutions in the face of climate change. Traditional crop selection methods rely on manual soil testing and historical knowledge, which can lead to high margins of error. These methods do not account for variations in soil composition, pest prevalence, and local climate conditions, all of which can significantly affect crop yield. Furthermore, with limited access to advanced analytical tools, many farmers are unable to leverage modern scientific methods for optimal crop selection.

Climate change further complicates these issues, with the increasing frequency of extreme weather events affecting soil moisture, nutrient levels, and pest populations. Farmers need accurate, real-time information to make decisions that align with evolving environmental conditions.

Key Points:

Articulation of the Problem:

Traditional crop recommendation methods are insufficient for modern agricultural complexities. There is an urgent need for sustainable farming solutions in light of climate change.

Context and Significance:

Relying on manual soil testing and historical knowledge results in high margins of error and inefficiency. These traditional methods fail to consider critical variations in soil composition, pest prevalence, and local climate conditions.

The Gap:

Farmers lack access to advanced analytical tools and modern scientific methods for optimal crop selection. Traditional methods don't provide real-time, accurate information needed for effective decision-making in agriculture.

Impact of Not Addressing the Problem:

Without innovative solutions, farmers will continue to face reduced crop yields, inefficient resource use, and increased vulnerability to environmental changes.

1.3 Objectives

Developed a deep learning-based system that classifies soil types from satellite imagery. Utilize EfficientNet B7 to analyse satellite images and identify soil characteristics such as texture and fertility. Achieve high accuracy (99%) in soil type classification using advanced activation functions like Swish and Mish.

Crop Recommendation System: Implement a crop recommendation engine that suggests the most suitable crops based on the classified soil type. Consider additional factors such as regional climate, water availability, and historical crop performance when making recommendations. Provide real-time, personalized crop recommendations for farmers, ensuring they make the best decisions for maximizing yield.

User-Centric Interface: Design and develop an intuitive mobile application that allows farmers to easily register, input their location, and access the recommendations. Integrate Google Maps to show the user's land on a satellite view, helping them visualize their soil conditions and location. Enable farmers to track their crop recommendations and performance over time, helping them make data-driven decisions.

Integration of Satellite Imagery: Collect and utilize Sentinel Hub satellite images for soil type classification, providing up-to-date data on the land's characteristics. Preprocess and label satellite images to ensure they are suitable for machine learning model training.

Model Training and Optimization: Train the EfficientNet B7 model using a large dataset of satellite images to classify soil types accurately. Optimize the model's performance using state-of-the-art deep learning techniques and fine-tune hyperparameters to improve prediction accuracy.

Sustainability and Resource Optimization: Help farmers optimize their use of resources such as water by recommending crops that are well-suited to the local soil type and climate. Promote sustainable agricultural practices by reducing the reliance on trial-and-error methods, improving crop yields, and minimizing environmental impact.

Accuracy and Reliability: Strive for a high level of accuracy in both soil classification and crop recommendations (target of 99% accuracy). Provide transparency in prediction results, allowing farmers to see the model's confidence levels for soil classification and crop suitability.

User Education and Empowerment: Educate farmers about the benefits of precision agriculture and the science behind soil and crop relationships. Empower farmers with actionable insights and recommendations that can lead to better decision-making, enhanced productivity, and improved financial outcomes.

1.4 Scope of the Project

Scope of the project The CropIQ project focuses on the development and implementation of a precision agriculture system aimed at improving farming practices through technology. The scope of this project can be divided into several key areas that define its boundaries, objectives, and potential impact. Below are the primary aspects of the project's scope:

1. Target Audience

Farmers: The main users of the CropIQ platform will be farmers, especially those in rural areas, who can benefit from data-driven insights for crop selection.

Agricultural Advisors/Researchers: The platform can also be used by agricultural researchers or consultants who may want to analyze and recommend suitable crops based on soil type and climate data.

2. Geographic Coverage

Regional Focus: The initial phase of the project will focus on specific geographic regions where satellite data for soil and crop conditions is available, such as areas with diverse agricultural practices (e.g., states in India).

Future Expansion: In the long term, the system can be expanded globally to accommodate diverse climates, soils, and farming practices, integrating more satellite imagery sources and crop recommendation databases.

3. Data Utilization

Satellite Imagery: The project relies on Sentinel Hub satellite images to collect and analyze soil data. These images will be processed to classify soil types and conditions in specific regions.

Crop Recommendation Database: A comprehensive database of crop types, soil compatibility, regional agricultural data, and climate information will be used to provide suitable crop recommendations based on soil type.

User-Provided Data: Users will input their location (through GPS), and the platform will use this data in combination with satellite images to provide personalized crop recommendations.

4. Technological Implementation

Machine Learning: The project will implement deep learning models, specifically EfficientNet B7, to classify soil types from satellite imagery. Activation functions such as Swish and Mish will be employed to optimize the model's accuracy.

Mobile Application: The system will be integrated into a user-friendly mobile app, allowing farmers to easily register, access location-based data, and receive crop recommendations.

Google Maps Integration: The app will integrate with Google Maps to provide farmers with satellite views of their land, enhancing the user experience and enabling them to visualize their soil conditions and location.

Cloud-based System: A cloud infrastructure will be used for data storage, model training, and processing, ensuring scalability and efficiency.

5. Functional Scope

Soil Classification: Classify the soil into various categories based on satellite images, such as clay, sandy, or loamy soil.

Crop Recommendation: Provide a list of suitable crops for a given soil type and region, considering factors such as soil fertility, climate conditions, and regional agricultural trends.

User Engagement: Enable farmers to access personalized recommendations, track the performance of recommended crops, and adjust their farming practices accordingly.

6. Out of Scope

Soil Testing and Physical Sensors: The project will not involve direct soil testing or the use of physical sensors for in-field data collection. It relies entirely on satellite imagery and location-based data to assess soil conditions.

Large-Scale Crop Management: While CropIQ will recommend suitable crops, it will not manage the entire agricultural process, such as irrigation scheduling, pest control, or harvesting.

Real-time Weather Integration: Although weather data can be useful for farming decisions, it will not be integrated into the initial version of the platform, as the focus is primarily on soil and crop selection based on satellite imagery.

7. Impact and Deliverables

Improved Yield and Efficiency: The system aims to improve farming outcomes by providing farmers with accurate and personalized crop recommendations, helping them optimize their land and resource use.

Sustainability: By recommending crops suited to the soil and climate, the platform will promote sustainable farming practices, reducing overuse of fertilizers, water, and other resources.

Scalability: As the system evolves, it can be expanded to include additional data sources, more regions, and new features such as pest detection, disease monitoring, and irrigation management.

Future Integrations: In future phases, CropIQ could integrate with other technologies such as IoT sensors, weather data services, and machine vision systems for comprehensive farm management solutions.

8. Limitations

Data Dependency: The accuracy of the crop recommendations depends on the quality and availability of satellite images and the crop recommendation database. Any gaps in the data may affect the reliability of the system.

Geographic Limitations: Initially, the system will focus on regions where sufficient satellite imagery and agricultural data are available. This might limit its applicability in some rural or less-documented regions.

The CropIQ project is aimed at creating an innovative, data-driven system to assist farmers in making informed decisions regarding crop selection. By leveraging satellite imagery, machine learning models, and location-based services, it will optimize farming practices, improve crop yields, and contribute to sustainable agricultural development. The scope of the project encompasses the development of a mobile application for real-time, personalized recommendations, ensuring that the system has a meaningful impact on the agricultural sector. CropIQ aims to be a robust tool for improving crop yield and sustainability across varied agricultural landscapes.

1.5 Methodology Overview

The methodology for the CropIQ project is structured to ensure a systematic approach to problem-solving and achieving the desired outcomes. It integrates modern technological practices with machine learning, satellite imaging, and user-centric application design. Initially, data collection involves acquiring high-resolution satellite images from the Sentinel Hub API, focusing on specific regions for soil classification, with images categorized by geographical areas. Additionally, crop compatibility data, including soil types, crop growth conditions, and regional agricultural trends, is gathered in a structured CSV format. In the preprocessing stage, satellite images undergo normalization, noise and artifact removal, and resizing to fit the input dimensions required by EfficientNet B7, while crop data is cleaned and organized. For soil classification, the methodology employs EfficientNet B7, a state-of-the-art CNN architecture, enhanced by Swish and Mish activation functions, to achieve high accuracy.

The model is trained using labeled satellite images of various soil types and optimized through techniques like the AdamW optimizer and learning rate scheduling. Performance evaluation is conducted using metrics such as accuracy, precision, recall, and F1 score, along with cross-validation to improve generalization. The crop recommendation system maps soil types to crops based on the compiled dataset and additional factors like water availability, and historical crop yield data. A decision-making algorithm provides real-time crop recommendations by integrating with the trained soil classification model.

Future Works

The user-friendly Android application allows farmers to register, input their location, and view crop recommendations. Deployment involves setting up a pipeline to periodically retrain the model with new data to improve accuracy and reliability. Rigorous system testing ensures app functionality across different devices and regions, while model validation involves expert reviews and real-world data comparison. User testing engages farmers to refine the app based on their feedback. Sustainability is promoted by suggesting resource-efficient crops, and future enhancements include integrating weather data and IoT-based soil health monitoring.

CHAPTER-II

LITERATURE REVIEW

This chapter provides an in-depth exploration of existing research, technologies, and methodologies that inform and support the development of CropIQ. The literature review addresses recent advancements in agricultural technology, remote sensing, and machine learning applications for precision farming, offering context and insights that shape CropIQ's approach.

2.1 Overview of Related Work:

This section introduces the evolution of crop recommendation systems, particularly those enhanced by remote sensing and machine learning. It highlights how traditional methods have been improved through satellite imagery, enabling real-time monitoring of soil and vegetation. Studies focused on remote sensing, such as the use of Sentinel-2 data, illustrate the potential for analysing soil moisture.

This foundational research shows how satellite images can reveal critical agricultural insights, such as detecting crop, soil composition, and environmental changes, all of which are central to CropIQ. By building on these advancements, CropIQ leverages satellite imagery and machine learning to provide precise, data-driven crop recommendations, addressing the limitations of traditional methods and promoting sustainable agriculture practices.

2.2 Review of Similar Projects and Research Papers:

Satellite- and drone-based remote sensing of crops and soils for smart farming- [1]

Author- Yoshio Inoue, Journal: Soil Science and Plant Nutrition

Remote sensing technologies have revolutionized the way modern agriculture is practiced, especially with the advent of satellite images. Inoue (2020) highlights the critical role these technologies play in smart farming (SF), enabling more efficient, sustainable and precise crop production by integrating sensing, robotic and information communication technologies.

Recent advances in remote sensing technologies include the deployment of multispectral and hyperspectral sensors on satellites. These systems capture high-resolution data across a range of wavebands, providing detailed

information on crop growth and soil fertility.

Machine Learning for Precision Agriculture Using Imagery from Unmanned Aerial Vehicles (UAVs): [2]

Author- Imran Zulkarnaen, Daa Addeen Abuhani Maya Haj Hussain.

Journal: Drones

The main objective of this paper is to survey the application of machine learning techniques, particularly convolutional neural networks (CNNs), combined with UAV imagery in the field of precision agriculture. The goal is to improve crop monitoring by solving complex agricultural tasks like crop classification, weed detection, cropland mapping, and field segmentation by using machine learning and deep learning models on UAV- acquired images, the aim is to increase productivity, improve yield, and utilize resources effectively. This study concludes by comparing the effectiveness of various machine learning methods against current traditional practices to demonstrate the potential benefits for farmers and stakeholders

Deep Learning-Based Classification of Remote Sensing Images: Challenges, Techniques, and Future Directions in Global Sustainability [3]

Author- Biplov Paneru, Bishwash Paneru, Sanjog Chhetri Sapkota.

Journal- Aerospace Engineering

The images are preprocessed to 256x256 pixels, then split into a training and validation set. The models are trained using the Adam optimizer and evaluated on accuracy and validation metrics to determine the effectiveness in enhancing sustainable decision-making for agriculture, environmental monitoring, and resource management the study demonstrates the pros and cons of each model for RS image classification, while. MobileNetV2 faced significant overfitting issues.

An Intelligent Crop Recommendation System using Deep Learning [4]

Author- Dr. Kanaga Suba Raja S, Dr. V. Balaji,

The study is based on the paper "An Intelligent Crop Recommendation System using Deep Learning" which aims to develop a robust system that gives accurate crop recommendations by means of deep learning techniques. Their intention is to refine typical crop selection processes using deep neural networks. It uses different agriculture information, historical crop performance, climate variables and soil conditions through the development of precision agriculture and it is trying to encourage farmers take up more sustainable production types by suggesting crops that make most efficient use of resources and rain with

minimum environmental impact. It aims to substantiate its efficacy at farm sites — so that farmers can use state-of-the-art instruments for enhanced production and sustainable agriculture.

Machine learning (ML) in agriculture

The main purpose of this paper is to discuss the applications of machine learning (ML) in agriculture specifically focusing on disease detection, crop production prediction and soil parameter prediction. IT also includes smart irrigation and livestock production for Machine Learning. This was intended purpose to show off the potential for ML in sustainable agriculture

The recognition of crops with spectral and temporal information

The work has as objective to improve the recognition of crops with spectral and temporal information, for which convolutional neural networks (CNNs) were used combined synthetic images that join different moments and spectral data. The focus is on agricultural monitoring as it will increase productivity, yield of the crop and utilize resources properly. This method will be concluded with comparison of the potential benefits going to farmers and other commercial stakeholders against present use practices

2.3 Summary and Gap Identification:

Remote sensing and machine learning technologies have significantly advanced modern agriculture, providing tools for more efficient, sustainable, and precise crop production. Inoue (2020) discusses the critical role of satellite-based remote sensing in smart farming, highlighting how multispectral and hyperspectral sensors capture high-resolution data that offer detailed insights into soil and crop health. Zulkarnaen et al. focus on the use of machine learning techniques, particularly convolutional neural networks (CNNs), combined with UAV imagery to enhance crop monitoring, classification, weed detection, and field segmentation.

Their study demonstrates the advantages of these technologies over traditional practices. Paneruet al. address the challenges and techniques of deep learning-based classification of remote sensing images, emphasizing the importance of preprocessing for accuracy. They work on an intelligent crop recommendation system leverages deep learning to refine crop selection processes, incorporating diverse agricultural data to promote sustainability. Additionally, other studies highlight the potential of machine learning in agriculture for tasks such as disease detection, crop production prediction, and smart irrigation, showcasing its transformative impact.

Gap Identification:

Despite the advancements in remote sensing and machine learning for agriculture, several gaps remain:

Data Integration: There is a need for systems that integrate comprehensive, real-time data from multiple sources to enhance the accuracy and reliability of crop recommendations.

Model Robustness: Existing models often lack robustness and struggle with generalization due to the variability in environmental conditions and the need for continuous data updates.

User Accessibility: There is a gap in developing user-friendly interfaces that make advanced agricultural technologies accessible to farmers with varying levels of technical expertise.

Sustainability Focus: Current solutions do not adequately address the need for sustainable farming practices, especially in the context of climate change and resource limitations.

The CropIQ project aims to bridge these gaps by leveraging advanced satellite imagery and machine learning to provide precise, region-specific crop recommendations. The project focuses on integrating real-time data, improving model robustness through continuous updates, creating an accessible user interface, and promoting sustainable agricultural practices.

CHAPTER-III

SYSTEM ANALYSIS

3.1 Requirements Gathering

The main goal of the crop recommendation system is to assist farmers in selecting the most suitable crops for specific soil types by analyzing satellite imagery. The system collects and processes satellite data (Sentinel-2 images), soil information, and climate variables. The gathered data is pre-processed and fed into a deep learning model (EfficientNet-based CNN) to make crop recommendations. These recommendations help farmers optimize crop yield and sustainability by selecting crops aligned with their local soil conditions.

3.2 Functional Requirements

These are the core features the system must provide to fulfil the project goals:

- 1. User Management:**
 - a. Allow registration with username, password, location, and language preference.
 - b. Enable secure login/logout functionality.
- 2. Location-Based Soil Type Analysis:**
 - a. Retrieve user location using GPS to determine soil type.
 - b. Integrate with Google Maps to provide a satellite view of the user's land.
- 3. Crop Recommendation System:**
 - a. Analyse soil type and region to recommend suitable crops.
 - b. Display crop suggestions with accuracy levels in an easy-to-read format (e.g., images).
- 4. Model Integration:**
 - a. Efficient Net B7 model for soil classification with high accuracy.
 - b. Use Swish and Mish activation functions for improved performance.
- 5. User Interface:**
 - a. Design a simple, user-friendly Android application interface for farmers.

3.3 Non-Functional Requirements

These ensure the system is reliable, secure, and scalable:

1. **Performance:**
 - a. Achieve 99% accuracy in soil classification and crop recommendation.
 - b. Provide real-time responses for crop suggestions.
2. **Scalability:**
 - a. Handle many users and expand to cover additional states or regions.
3. **Usability:**
 - a. Ensure the app is easy to navigate, even for users with minimal technical knowledge.
 - b. Provide support for multiple languages to cater to diverse users.
4. **Reliability:**
 - a. Maintain uptime and ensure smooth functioning without crashes.
5. **Security:**
 - a. Protect user data with secure storage and encrypted communication.
 - b. Ensure location access is used solely for the intended purpose.
6. **Portability:**
 - a. Make the app compatible with various Android versions and devices.

3.4 Feasibility Study

3.4.1 Technical Feasibility

Technical feasibility assesses whether the project can be executed with the current technology and resources. It involves evaluating the technical requirements, available tools, expertise, and potential challenges. For CropIQ, this would include examining the capabilities of satellite imagery (e.g., Sentinel-2) and machine learning models (e.g., EfficientNet). It also involves checking the availability of necessary software and hardware, such as high-performance computing for processing large datasets and robust infrastructure for deploying the crop recommendation system. Technical feasibility ensures that the project can be technically implemented and that any technical risks can be managed effectively.

3.4.2 Operational Feasibility

Operational feasibility assesses whether the project can be integrated into the existing operations and how it will impact current processes. It evaluates whether the organization has the operational capacity to support the project, including the availability of skilled personnel and the ability to manage changes in workflow. For CropIQ, this involves determining how the system will be adopted by farmers, ensuring that they can easily use the technology, and assessing the support required for training and ongoing maintenance. Operational feasibility ensures that the project will function smoothly within the current operational framework and meet the needs of its users.

3.4.3 Economic Feasibility

Economic feasibility, also known as cost-benefit analysis, evaluates whether the project is financially viable. It involves estimating the costs of developing, implementing, and maintaining the project compared to the expected benefits. For CropIQ, this includes costs related to acquiring satellite images, developing machine learning models, and creating the user application. It also considers potential funding sources, expected returns from increased crop yields, and cost savings from optimized resource use. Economic feasibility aims to ensure that the project will provide a good return on investment and be financially sustainable in the long run.

3.4.4 Risk Analysis

Risks include potential inaccuracies in satellite data, model overfitting, and climate variability affecting predictions. Steps such as continuous model updates, rigorous data validation, and incorporating real-time weather data can mitigate these risks

CHAPTER -IV

SYSTEM DESIGN

4.1 Overall System Architecture

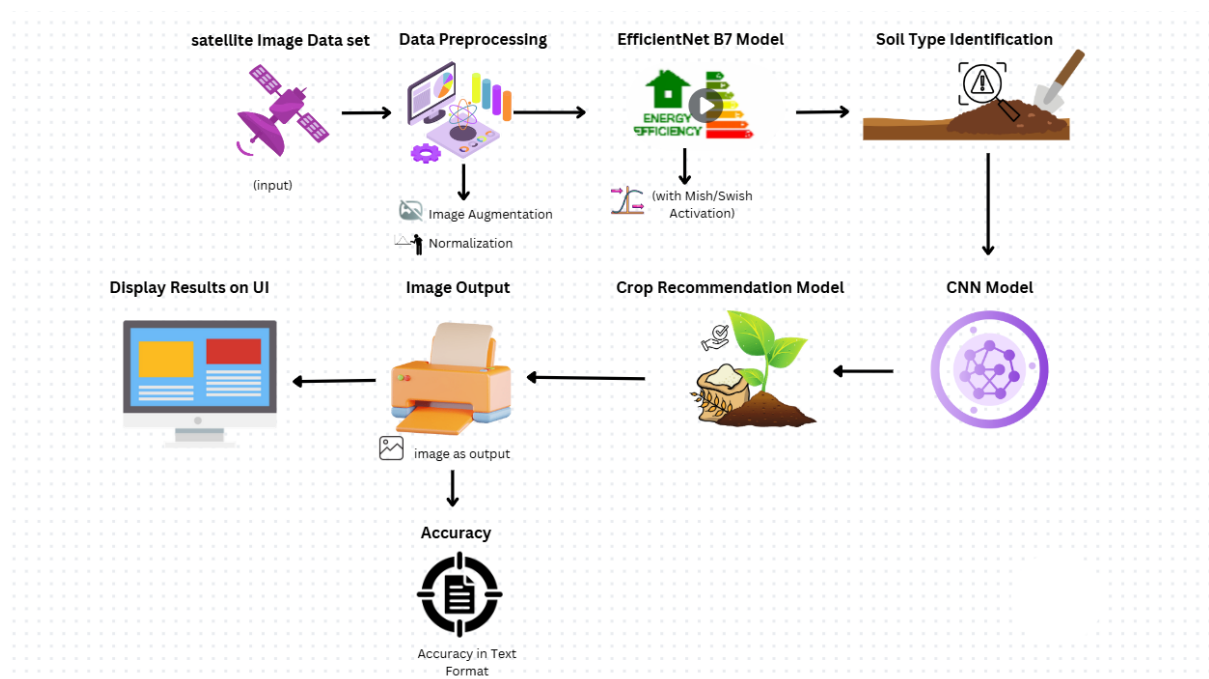


Fig-1 Proposed System of CropIQ Model

4.2 Module Design

To keep the CropIQ system as concise and efficient as possible, we can focus on key data that directly contributes to the effectiveness of crop recommendations. Below is a list of the most important data types and their respective purpose in the system:

4. Satellite Imagery Data (Sentinel-2)

- Data Type: Multispectral images (including bands like Red, Green, Blue, NIR, SWIR).

- Purpose: To assess vegetation health, soil moisture, and overall environmental conditions. Provides the primary input for the model.

2. Crop Data (Targeted)

- Data Type: Historical data of crop yields, crop types, and success rates in various regions.
- Purpose: To refine the crop recommendation model based on past performance in specific regions.

3. Geographic Location

- Data Type: Latitude, longitude, region identifier.
- Purpose: To map satellite images and contextualize recommendations based on the farmer's location.

4.2.1 Module 1

Data Collection Module: -

Objective:

- To gather the necessary satellite imagery and related metadata for crop recommendation analysis.

Key Features:

- **Satellite Image Retrieval:**

Access high-resolution Sentinel-2 imagery through an API (e.g., Copernicus Open Access Hub).

Image selection: Filter by geographical region and time period.

- **Metadata Collection:**

Extract key metadata for each image, such as acquisition time, cloud cover, and geographical location.

- **Data Storage:**

Store images and metadata in cloud storage or a local database.

Input Data:

- **Location:** Latitude and longitude coordinates of the region.
- **Time Period:** Date range for which satellite images are requested.
- **Area of Interest:** Specific geographic regions or coordinates where satellite images are needed.

Output Data:

- **Satellite Images:** Raw multispectral data from Sentinel-2.
- **Metadata:** Details about the images such as acquisition time, location, and cloud cover percentage.

4.2.2 Module 2(Repeat as per your requirements)**Data Preprocessing Module:****Objective:**

To preprocess satellite imagery for accurate analysis.

Key Features:

- **Atmospheric Correction:** Adjust for atmospheric distortions using tools like sentinelsat or pyresample.
- **Geometric Correction:** Align satellite images to a common coordinate system.
- **Radiometric Calibration:** Convert raw image data to reflectance values.
- **Noise Removal:** Apply filters to reduce noise and enhance image quality.

Technology Stack:

- Python libraries: opencv, Tensorflow, matplotlib, numpy, pandas, seaborn.
- This module ensures that satellite images are clean, calibrated, and ready for feature extraction.

Machine Learning

- **Model Architecture:** EfficientNet B7
- **Activation Functions:** Swish and Mish

- **Frameworks/Libraries:**

- TensorFlow/Keras (for model development and training)
- Scikit-learn (for preprocessing and metrics evaluation)
- OpenCV (for image processing)

- **Dataset Management:**

- Satellite Images (from Sentinel Hub)
- CSV File (for crop recommendations by region)

4. Tables Overview

- Images: Stores satellite image metadata.
- Soil_Data: Stores soil characteristics for regions.
- Crop_Recommendations: Stores the output of crop recommendations.

3. Relationships:

- Images to Soil_Data: One-to-one relationship (by location_id).
- Images to Vegetation_Indices: One-to-many relationship (by image_id).
- Soil_Data to Crop_Recommendations: One-to-many relationship (by location_id).

-

4.3.1 ER Diagram

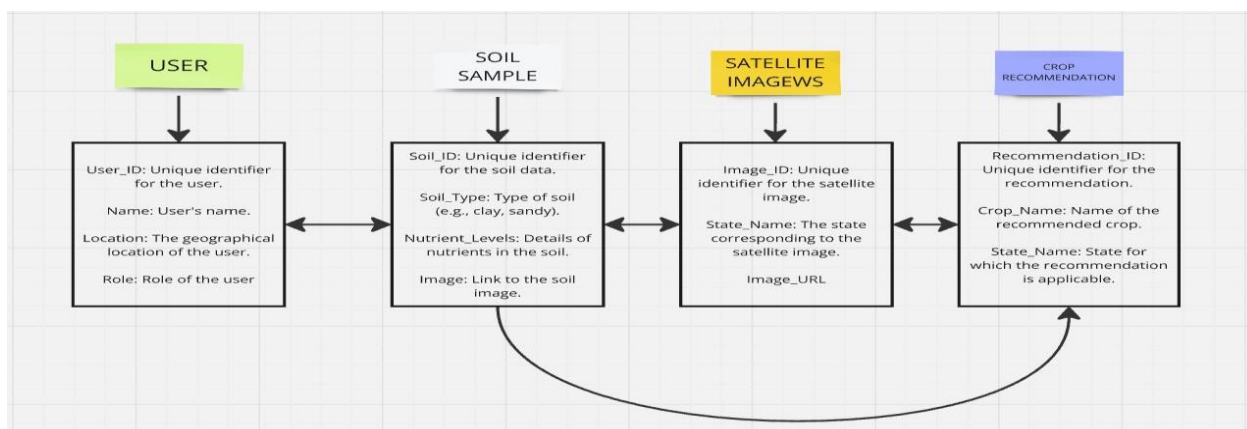


Fig-2 Entity Relation Diagram

Entity Relationship Diagram (ERD) for the CropIQ project. It includes the entities and relationships between Images, Soil_Data, Vegetation_Indices, and Crop_Recommendations. The diagram illustrates the one-to-one and one-to-many relationships between these entities

4.4 User Interface Design

User Interface Design for CropIQ

To make the CropIQ interface as straightforward as possible, we'll focus on three main sections: Dashboard, Map View, and Crop Recommendations. Each section will provide only essential data and functionality.

User Actions:

View Data for Current Date: Shows only the most recent data, without historical trends.

Refresh: Update to see the latest satellite data.

1. Crop Recommendations

Key Data Displayed:

Recommended Crop: Display the top recommended crop only.

- Confidence Score: A single score (e.g., 85%) showing recommendation accuracy.

User Actions:

- Download Recommendation: Export a simple PDF with the crop suggestion.
- Refresh Recommendation: Updates based on the latest soil and satellite data.

Design Style:

- Minimal Layout: Clean, with only key icons and labels.
- Simple Color Coding: Use green, yellow, and red to easily convey data status.
- Mobile-Friendly: Optimized for quick access on small screens.

- This minimal interface ensures that users receive essential crop recommendations and soil insights without unnecessary complexity.

4.4 User Interface Design

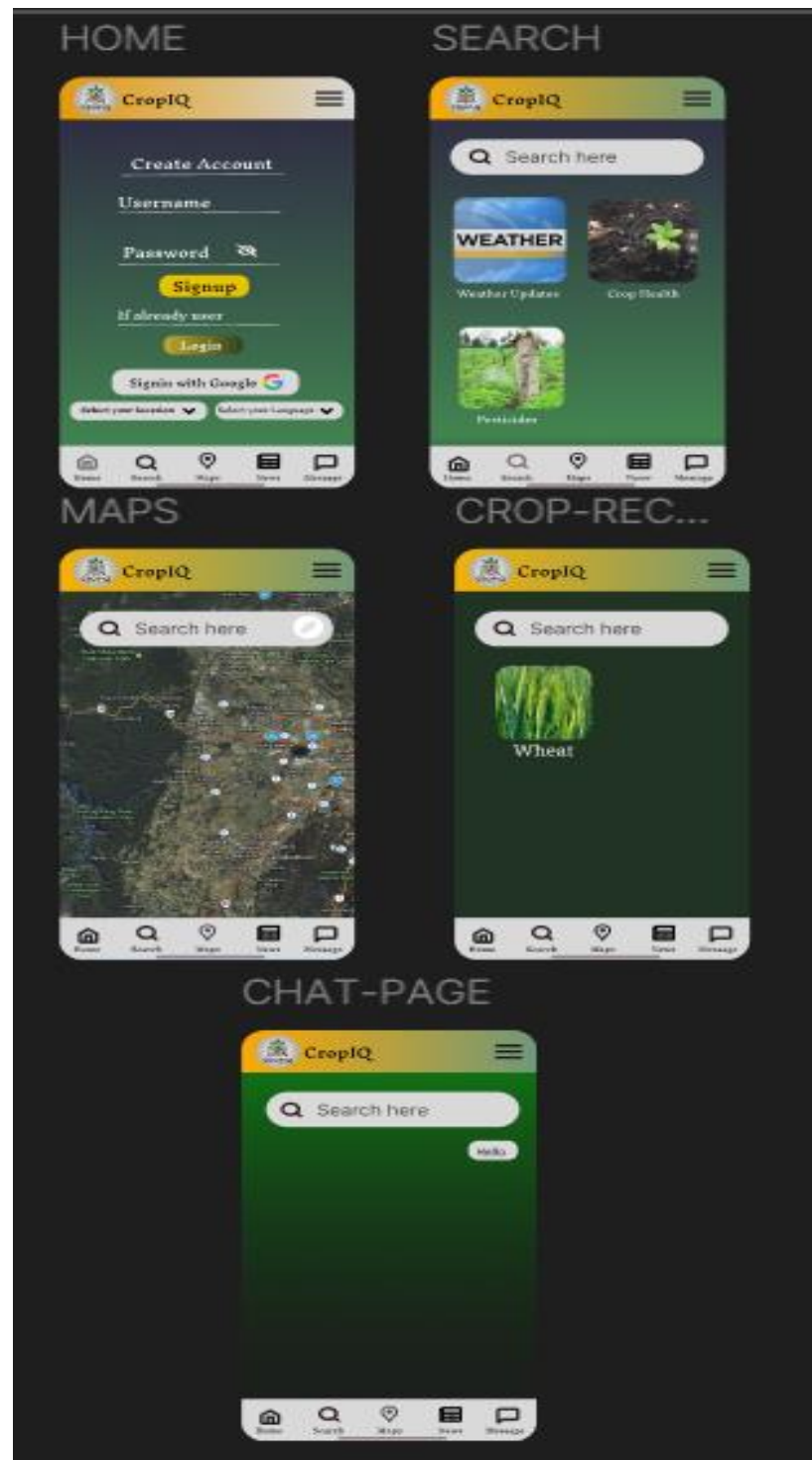


Fig-3 CropIQ-User Interface

1. Login/Registration Screen

Purpose: First point of interaction for the user.

Features:

- Input fields for:
 - Username
 - Password
- Buttons for:
 - Login
 - Register (redirects to the registration page)
- **Optional:** Social login options (Google, Facebook) for convenience.
- Design:
 - Background: Subtle gradient with agricultural imagery (e.g., fields or crops).
 - Buttons: Large, rounded corners with contrasting colors (e.g., green for login).
 - Language Dropdown: Top-right corner to select the user's preferred language.

2. Registration Screen

Purpose: Allows new users to sign up.

Features:

- Input fields for:
 - Full Name
 - Mobile Number
 - Email Address
 - Password (with confirmation)
 - State/City dropdown
 - User Role (Farmer, Student, etc.)
- Buttons for "Submit" and "Cancel."
- Checkbox for terms and conditions.

3. Home Screen

Purpose: Central dashboard for app features.

Sections:

- **Welcome Banner:** Greet the user by name.
- **Buttons/Icons:**
 - **Crop Recommendation:** Large button leading to soil/crop recommendation.
 - **Weather Update:** Shows a small card with the day's forecast.
 - **Satellite View:** Access Google Maps in satellite mode.
 - **Profile:** Icon to manage user details or app settings.
- **Navigation Bar** (at the bottom):
 - Home | Recommendations | Weather | Profile
- **Design:**
 - Clean layout with cards for features.
 - Use icons with labels for clarity (e.g., a plant icon for crop recommendations).

4. Crop Recommendation Screen

Purpose: Displays crop suggestions based on soil type.

Features:

- **Location Access:**
 - Button to allow/restrict location access.
- **Soil Information:**
 - Show soil type (based on location or satellite data).
- **Crop Recommendations:**
 - List of crops suitable for the soil type with:
 - Crop Name
 - Estimated Yield (optional)
 - Icons or images of crops.
- **Refresh Button:** To update location or soil data.

5. Satellite View Screen

Purpose: Display the user's location on Google Maps (satellite mode).

Features:

- Map centered on user's location.
- Overlays showing:
 - Soil type regions.
 - Crop zones (if available).
- Button to switch between "Normal" and "Satellite" view modes.

6. Weather Forecast Screen

Purpose: Provides localized weather updates.

Features:

- Current weather conditions:
 - Temperature, Rain Probability, Wind Speed.
- Weekly forecast in a scrollable format.
- Notifications:
 - Alerts for extreme weather (e.g., storms, heavy rain).

7. Profile/Settings Screen

Purpose: Manage user preferences.

Features:

- Update personal details.
- Change password.
- Manage language preference.
- Option to enable/disable notifications.
- Logout button.

General Design Principles

- **Typography:** Use clear and legible fonts like **Roboto** or **Poppins**.
- **Color Palette:**
 - Green (symbolizing growth) for buttons and accents.
 - Neutral background colors (light beige, white, or pastel shades).
 - Contrast for text (dark colors for readability).
- **Accessibility:**
 - Large buttons for easy tapping.
 - Support for regional languages.
 - Minimal text input; use dropdowns or radio buttons where possible.

Example Flow

1. **Login → Home Screen → Crop Recommendation Screen → Weather Forecast → Profile Settings.**
2. Users can navigate between sections using the bottom navigation bar or swipe gestures.

4.4.1 User Flow Diagram:

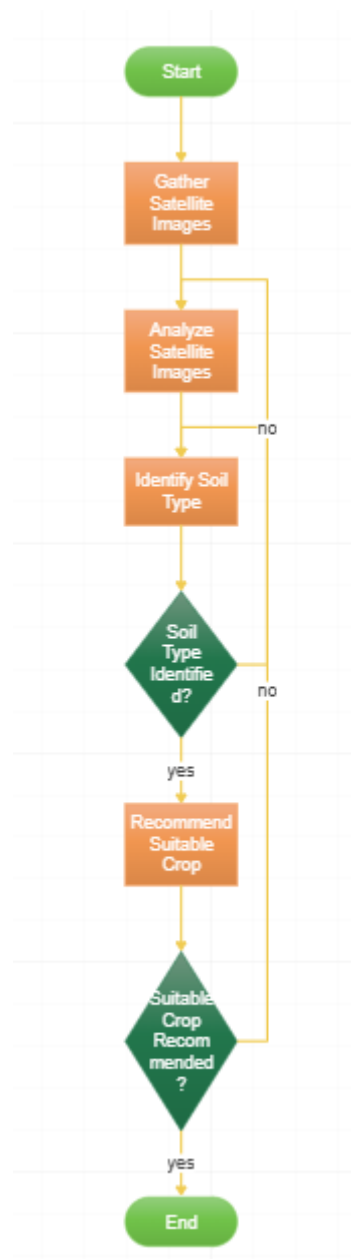


Fig-4 User Flow Diagram

Here is the digital user flow diagram for the CropIQ application. It visually represents the sequence of user actions, from logging in to exporting crop recommendations. Each step is displayed with clear flow arrows, making the user journey easy to follow.

CHAPTER -V

IMPLEMENTATION

5.1 Technology Stack

The crop recommendation system relies on a robust and scalable technology stack designed to handle large datasets and advanced machine learning processes. The primary components of the technology stack include:

Programming Languages: Python is the primary language used for this project due to its extensive libraries for data science, machine learning, and deep learning. Python's versatility allows for seamless integration of data preprocessing, model training, and deployment tasks.

Frameworks and Libraries: Key frameworks such as TensorFlow and Keras are used to implement the deep learning model. TensorFlow is highly efficient for processing large datasets, while Keras offers a user-friendly API to build and fine-tune neural networks.

Satellite Data Processing: The system utilizes Sentinel-2 satellite imagery, which is accessed through APIs provided by the Sentinel Hub. OpenCV and GDAL libraries are used for image processing, including noise removal, geometric correction, and feature extraction.

Modeling and Analysis Tools: For the deep learning model, the EfficientNet architecture is employed. This model is optimized for high performance on large datasets while keeping computational costs low. Additionally, the Swish and Mish activation functions are integrated to enhance model convergence and learning capabilities.

5.1.1 Programming Languages and Tools

Programming Languages: Python

Libraries and Frameworks: TensorFlow, Keras, OpenCV, GDAL, Pandas, NumPy, TPU (Tensor Processing Unit)

Satellite Data Source: Sentinel-2 (via Sentinel Hub API)

Visualization: Matplotlib and Seaborn for generating graphs and visual outputs of results.

5.2 Implementation of Modules

The project is broken down into two core modules: Data Processing and Crop Recommendation. Each module addresses a critical aspect of the system and together form the basis of the crop recommendation engine.

5.2.1 Module 1: Data Processing

This module focuses on the acquisition and preprocessing of satellite images. The steps include:

Data Acquisition: Sentinel-2 satellite images are downloaded using the Sentinel Hub API. These images cover various regions and are selected based on the geographic coordinates and timeframes that match the growing seasons.

Preprocessing: The images are preprocessed to ensure accuracy. This includes atmospheric correction, radiometric calibration, and noise reduction. OpenCV and GDAL libraries are used to clean and prepare the images for analysis. The images are georeferenced to align them with the correct geographic coordinates and resampled to ensure uniform resolution across datasets.

Feature Extraction: Relevant features, such as soil composition, vegetation health, and moisture levels, are extracted from the satellite imagery. These features form the input data for the crop recommendation model.

5.2.2 Module 2: Crop Recommendation System

This module is responsible for analyzing the preprocessed data and providing crop recommendations. Key tasks include:

Model Training: The deep learning model, based on the EfficientNet architecture, is trained using the extracted features from the processed satellite images. Swish and Mish activation functions are employed to optimize the model's learning process.

Prediction and Recommendations: After training, the model predicts which crops are most suitable for each soil type and environmental condition in different regions. These predictions are based on the specific features extracted from satellite data, such as soil health, moisture levels, and other environmental factors.

5.3 Integration of Modules

The two modules are integrated to provide a seamless workflow. Once the satellite images are processed in **Module 1**, the extracted features are fed into **Module 2**. The crop recommendation system takes this data, runs predictions using the trained deep learning model, and outputs a list of suitable crops for the given soil conditions and geographical region.

The integration ensures that the system operates efficiently, allowing real-time data to be processed and recommendations generated on demand. Additionally, the integration provides farmers with an easy-to-use interface to input location details and receive actionable crop recommendations based on their region's specific environmental factors

CHAPTER-VI

TESTING

6.1 Testing Methodology

The testing process for the crop recommendation system is crucial to ensure the accuracy, reliability, and efficiency of the system. A comprehensive testing approach is employed, consisting of various levels of testing to validate different parts of the system. The main goal is to verify that the system performs well in real-world scenarios by providing accurate and actionable crop recommendations.

The testing methodology follows a structured approach, starting with unit testing for individual components, progressing to integration and system testing, and culminating in user acceptance testing to ensure the system meets user requirements.

6.1.1 Unit Testing

Unit testing involves testing the smallest, independent components of the system, typically individual functions or methods. Each part of the system, such as satellite data processing, feature extraction, and model prediction, is tested in isolation to ensure correct functionality.

Example Test: Testing the satellite image preprocessing module to ensure that noise reduction, atmospheric correction, and image resampling are performed accurately.

Tools Used: Python's unittest or pytest frameworks.

6.1.2 Integration Testing

Once individual units are validated, integration testing is performed to ensure that different modules of the system work together as expected. This includes testing the interaction between the data processing module and the crop recommendation system.

Example Test: Testing the integration of satellite image processing with the EfficientNet model. Ensure that the output from the preprocessing module (feature extraction) is correctly fed into the crop recommendation model.

Tools Used: Integration tests are manually scripted, with some automation using frameworks like pytest.

6.1.3 System Testing

System testing evaluates the system as a whole, ensuring that it functions correctly in its entirety. All modules (data acquisition, preprocessing, recommendation) are tested together in a real-time environment using the full dataset.

Example Test: The system is tested by processing satellite images from Sentinel-2 and generating crop recommendations for different regions. This ensures that all components function cohesively and produce accurate results.

Key Focus: Performance testing, accuracy of crop recommendations, and scalability.

6.1.4 User Acceptance Testing (UAT)

User acceptance testing ensures that the system meets the end users' requirements and expectations. In this stage, actual users, such as farmers or agricultural experts, interact with the system to validate its functionality, usability, and relevance in real-world scenarios.

Example Test: Farmers input location data to receive crop recommendations, and the accuracy of the recommendations is assessed based on local knowledge or expert validation.

Feedback Loop: User feedback is collected, and necessary adjustments are made to improve the user interface and overall system performance.

6.2 Test Cases and Results

Test cases are designed to cover all possible scenarios, including normal operation, edge cases, and system failures. Below are examples of key test cases and their expected results:

Test Case 1: Input satellite data for a known agricultural region.

Expected Result: The system processes the data and recommends crops that match the known soil types and environmental conditions.

Test Case 2: Introduce noise in satellite images and run the preprocessing module.

Expected Result: The preprocessing steps should remove noise, and the system should generate accurate crop recommendations.

Test Case 3: Run the system with data from multiple states simultaneously.

Expected Result: The system should handle large datasets efficiently without crashing and provide accurate recommendations for each state.

Results: These test cases should demonstrate that the system meets performance benchmarks, including accuracy in recommendations and efficient processing times.

6.3 Bug Tracking and Resolution

To manage bugs and issues found during testing, a bug tracking system is employed. Each issue is logged with details including the type of error, the affected module, the severity of the bug, and steps to reproduce the issue.

Tracking Tool: Tools like GitHub Issues are used to track bugs.

Bug Resolution Workflow:

Identify: Issues are identified during testing.

Assign: Bugs are assigned to the relevant team members for resolution.

Fix: The development team addresses the bugs, making necessary code changes.

Re-test: After the fix, the relevant test cases are re-run to ensure the bug has been resolved without introducing new issues.

CHAPTER -VII

RESULTS AND DISCUSSION

7.1 System Output Screenshots:

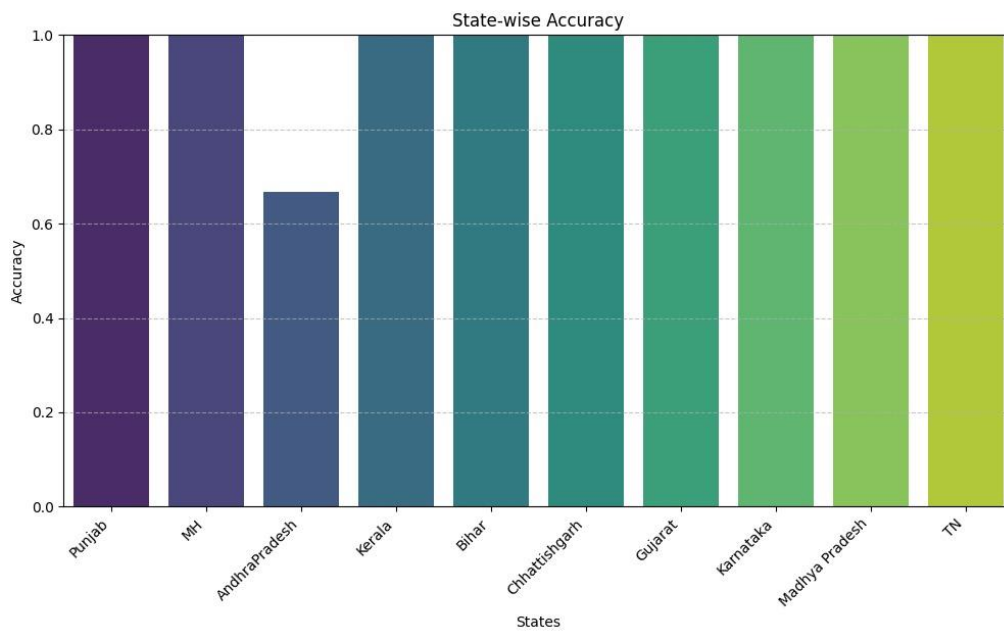


Fig-5 State Wise Accuracy Analysis

This bar graph visualizes the number of images or recommendations made for each state in your dataset.

Key Features:

X-axis: States (e.g., Andhra Pradesh, Bihar, Gujarat, etc.).

Y-axis: Count of satellite images or recommendations for each state.

Purpose:

- To understand the distribution of images or predictions across different states.
- To identify states with more or fewer samples, ensuring balanced training and testing.
- A high number of images for a state might mean more reliable recommendations.
- States with fewer images may require additional data collection.

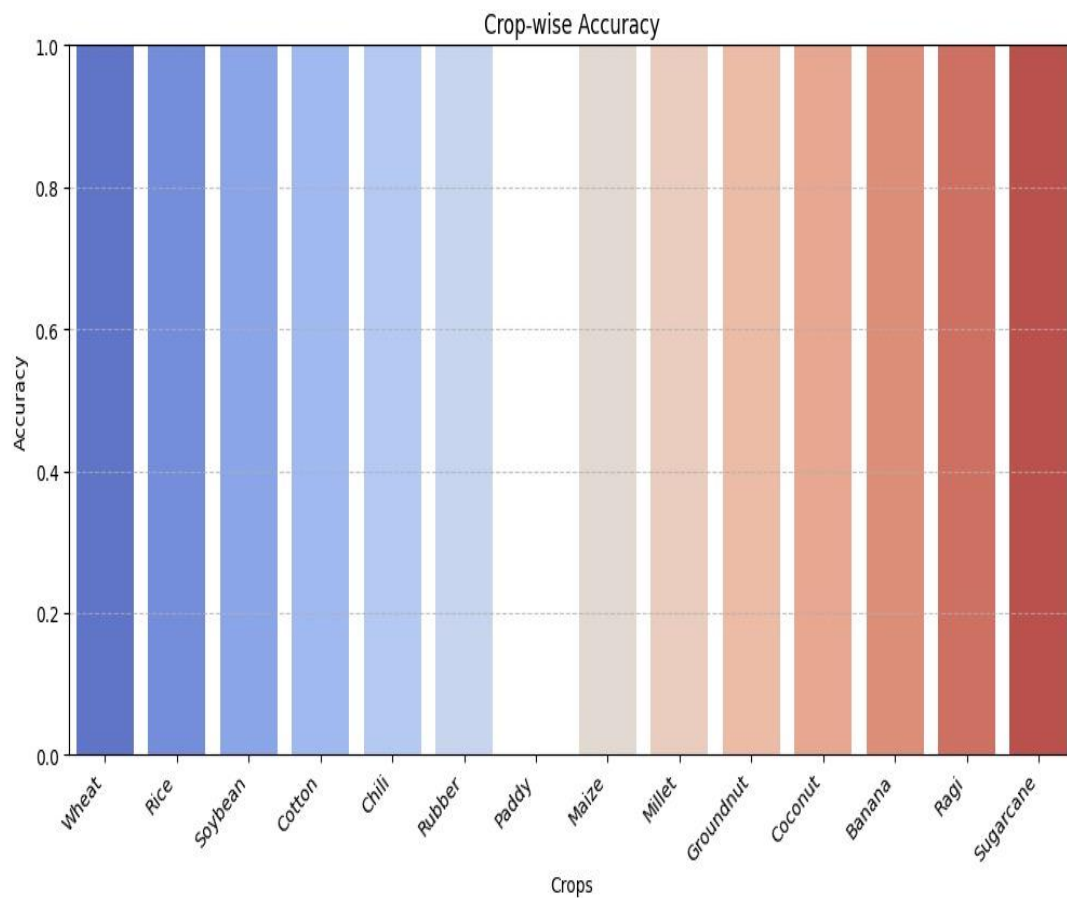


Fig-6 Crop Wise Accuracy Analysis

This graph represents the frequency or recommendations of different crops across the dataset.

Key Features:

X-axis: Crop names (e.g., Rice, Wheat, Sugarcane, etc.).

Y-axis: Count of occurrences or recommendations for each crop.

Purpose:

- To see which crops are most or least recommended.
- To evaluate whether the dataset or model is biased towards specific crops.
- A balanced crop distribution indicates the model covers various crop types.
- Skewed results might highlight dataset imbalance or model biases.

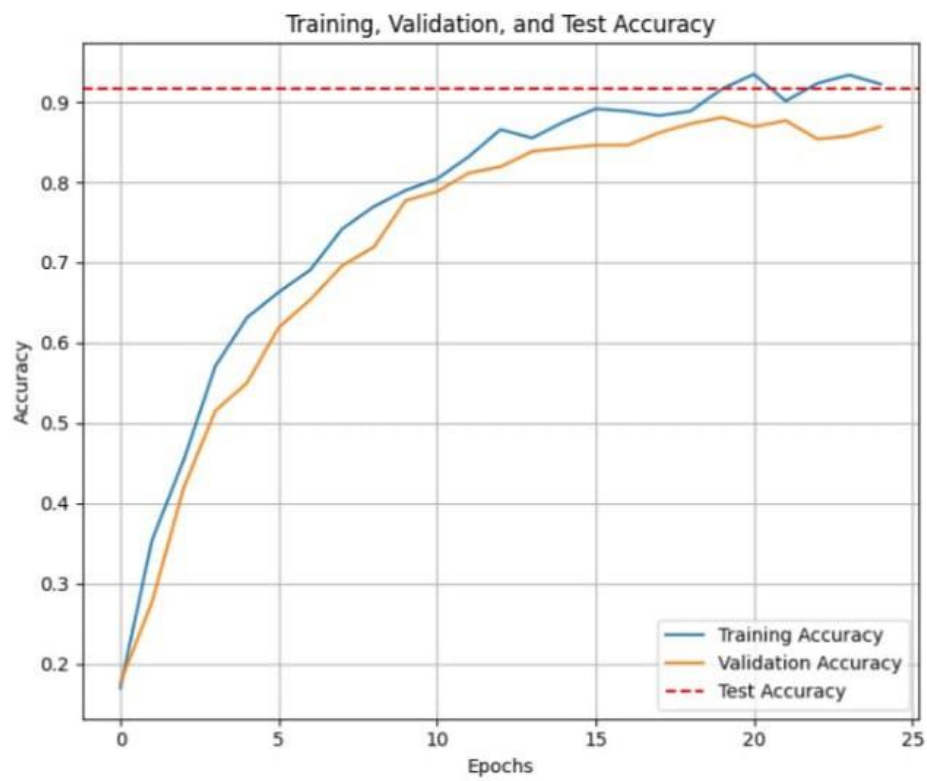


Fig-7 Training, Validation Accuracy

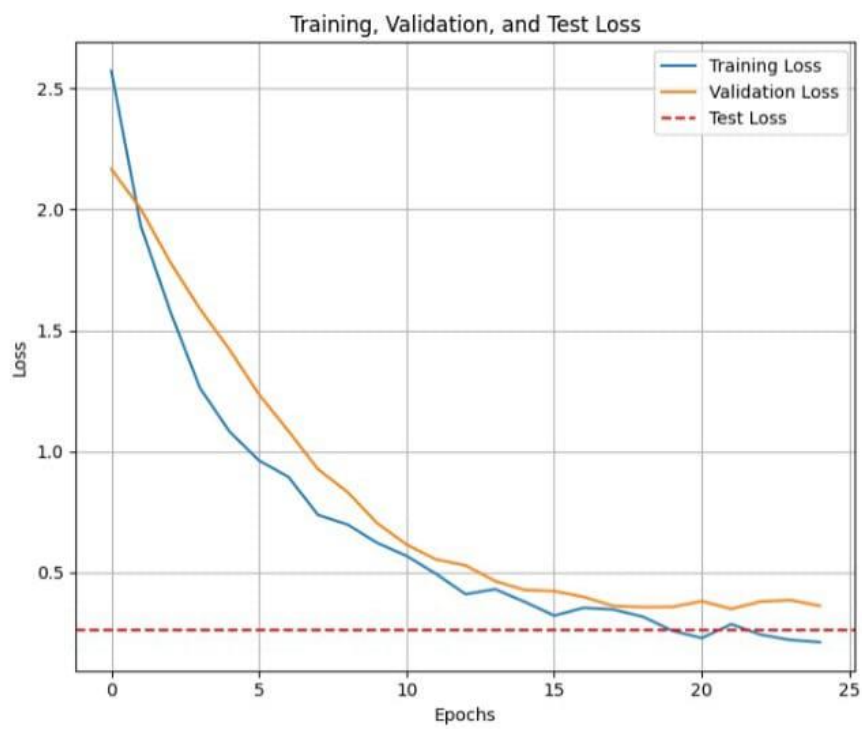


Fig-8 Training, Validation and Test Loss

Training and Validation Accuracy:

Accuracy measures the percentage of correct predictions the model makes during the training and validation phases.

Training Accuracy:

- Reflects how well the model fits the training dataset.
- High training accuracy indicates the model learns patterns in the training data.
- If accuracy is too high compared to validation accuracy, it may indicate overfitting.

Validation Accuracy:

- Reflects how well the model generalizes to unseen data.
- It is typically slightly lower than training accuracy.
- A consistent increase in validation accuracy across epochs shows the model's generalization ability is improving.

Training and Validation Loss:

Loss measures the error between predicted outputs and actual values.

Training Loss:

- Should decrease over epochs as the model optimizes its parameters.
- A very low training loss with high validation loss suggests overfitting.

Validation Loss:

- Should also decrease over epochs, but may plateau after a certain point.
- If validation loss increases while training loss decreases, the model is overfitting.

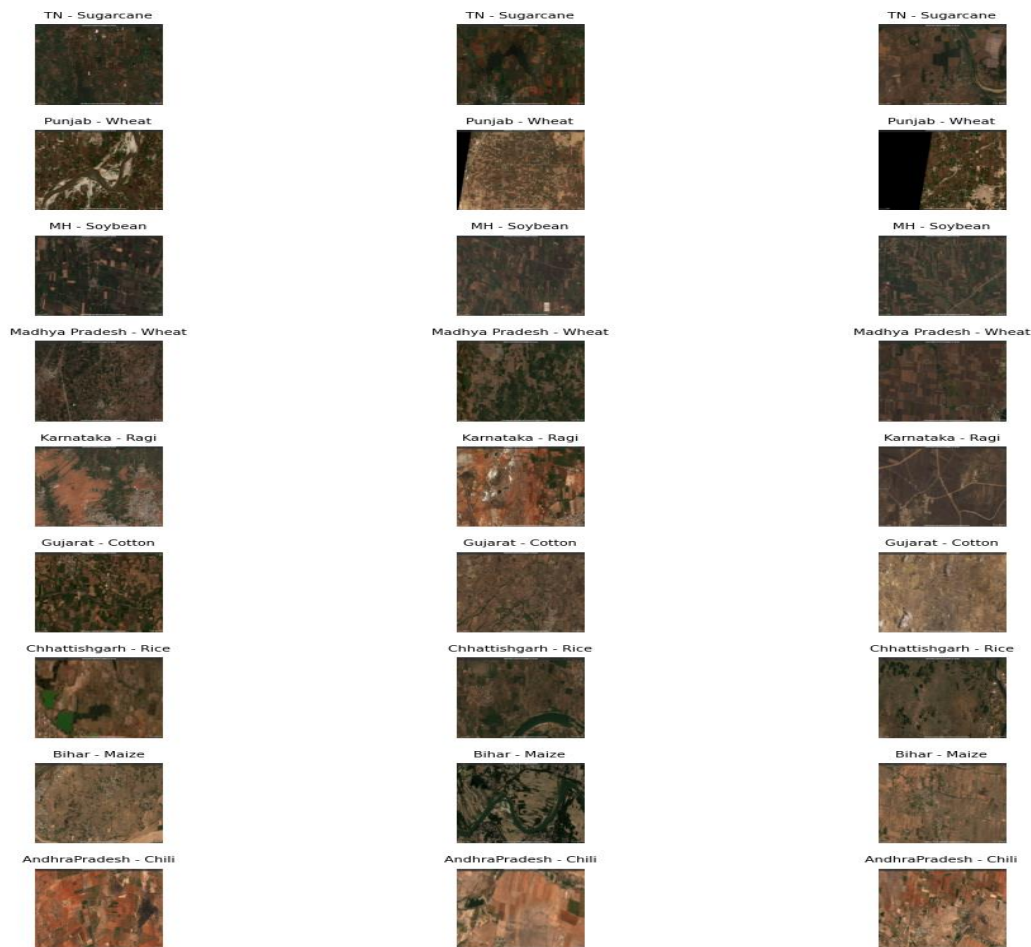


Fig-9 Proposed Crop Recommendation

In the CropIq Project demonstrate the results of the crop recommendation model. These images help visualize the model's predictions and are user-friendly, especially in applications like mobile apps or web interfaces. Here's how to approach the generation and presentation of output images:

Satellite Image:

The original satellite image acts as the base for predictions. It provides context to the user about the input data the model analyzed.

Predicted Crop Information:

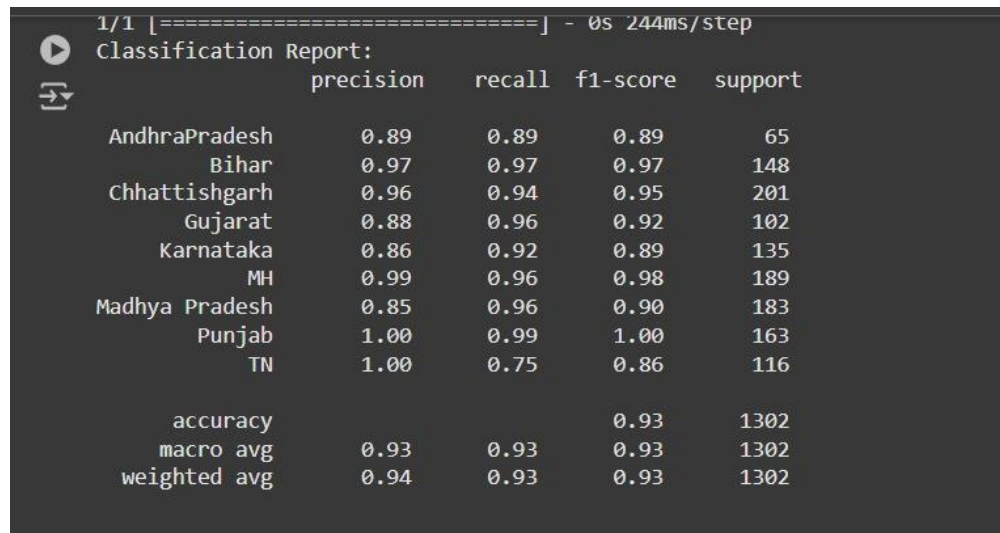
Crop Name: The name of the predicted crop (e.g., Rice, Wheat).

Confidence Score: Indicates how confident the model is about the prediction (e.g., 85% confidence).

Annotated Output:

Overlay the predicted crop name and confidence score directly on the satellite image. Optionally include additional details like the state name or alternative crop recommendations.

7.2 Evaluation Metrics



```

1/1 [=====] - 0s 244ms/step
Classification Report:

```

	precision	recall	f1-score	support
AndhraPradesh	0.89	0.89	0.89	65
Bihar	0.97	0.97	0.97	148
Chhattishgarh	0.96	0.94	0.95	201
Gujarat	0.88	0.96	0.92	102
Karnataka	0.86	0.92	0.89	135
MH	0.99	0.96	0.98	189
Madhya Pradesh	0.85	0.96	0.90	183
Punjab	1.00	0.99	1.00	163
TN	1.00	0.75	0.86	116
accuracy			0.93	1302
macro avg	0.93	0.93	0.93	1302
weighted avg	0.94	0.93	0.93	1302

Fig-9 Evaluation metrics

Classification Metrics for Each State:

The table lists eight states (e.g., Andhra Pradesh, Bihar, etc.) and their respective evaluation metrics:

Precision:

Measures the percentage of correctly predicted samples for each state out of all samples predicted as that state.

High Precision: Indicates fewer false positives.

Example:

Punjab has 1.00 precision, meaning all predictions for Punjab were correct. Karnataka has 0.86, meaning some incorrect predictions were made.

Recall

- Measures the percentage of correctly predicted samples for each state out of all actual samples from that state.
- High Recall: Indicates fewer false negatives.

Example:

- Gujarat and Bihar have 0.96 and 0.97 recall, showing the model correctly identified most samples for these states.
- Tamil Nadu (TN) has 0.75, indicating the model missed some TN samples.

F1-Score

The harmonic mean of precision and recall, balancing the trade-off between the two.

Example:

Madhya Pradesh has an F1-score of 0.90, showing a balanced performance.

Tamil Nadu has 0.86, slightly lower due to the lower recall.

Overall Performance

At the bottom of the table, metrics are averaged across all states:

Accuracy

0.93 (93%): Indicates the overall proportion of correctly classified images out of the total dataset.

Macro Average

Definition: The unweighted average of precision, recall, and F1-score across all states.

Value: 0.93, indicating that each state contributes equally to the average regardless of its number of samples.

Weighted Average

Definition: The weighted average of precision, recall, and F1-score, taking into account the support (number of samples) for each state.

Value: 0.94, slightly higher than the macro average because states with more samples contribute more heavily.

7.3 Comparison with Existing Systems:**CropIQ Improves on Existing Systems**

- CropIQ distinguishes itself from traditional and modern agricultural systems through several key improvements

1. Data Source:

- Traditional systems rely on manual soil testing and historical knowledge, while CropIQ uses high-resolution satellite data (Sentinel-2) to analyze soil and vegetation characteristics, offering deeper insights.

2. Real-time Analysis:

- Unlike systems that only provide periodic updates or rely on older data, CropIQ enables real-time insights on soil and crop health, helping farmers make timely decisions that respond to current conditions..

3. Precision in Crop Recommendations:

- CropIQ uses advanced machine learning (EfficientNet CNN) for accurate, location-specific recommendations, improving on existing systems that often provide broader, less tailored advice.

4. User Accessibility:

- Many systems require specialized knowledge or onsite consultations, but CropIQ is designed for ease of use with a mobile-friendly interface, making advanced insights accessible to more users.

5. Environmental Adaptation:

- Traditional methods and even some modern systems lack adaptability to climate change impacts, but CropIQ dynamically adjusts recommendations based on real-time environmental data, improving resilience.

6. Sustainability Focus:

- CropIQ prioritizes sustainable practices by recommending crops that align with local soil and climate conditions, reducing resource usage and promoting environmental health.

7. Customization and Specificity:

- While other systems may generalize recommendations across larger regions, CropIQ offers fine-grained insights for specific areas within a field, allowing farmers to address unique soil conditions accurately.

CropIQ offers a data-rich, real-time, and sustainable approach to crop recommendation, designed for ease of use and adaptability, setting it apart from traditional and existing systems.

7.4 Challenges Faced

The CropIQ project, while promising, comes with several challenges that must be addressed to ensure success:

1. Data Quality and Availability:

- High-resolution satellite data, like Sentinel-2, may have inconsistencies due to cloud cover, weather conditions, or technical limitations. Ensuring high-quality, cloud-free images is essential for accurate soil and vegetation analysis.

2. Complex Data Processing:

- Satellite data requires extensive preprocessing (e.g., atmospheric correction and noise reduction) to make it suitable for analysis. This is a computationally intensive process, and even minor errors in preprocessing can affect the model's accuracy.

3. Model Complexity and Training:

- CropIQ uses deep learning models, which require large amounts of data and computational power to train. Ensuring the model is accurate, especially in diverse soil and climate conditions, is a challenging task.

4. Limited Internet and Device Access for Users:

- Many farmers, especially in rural areas, may have limited access to high-speed internet or devices capable of running data-heavy applications. Designing a lightweight, mobile-friendly interface that still delivers insights is essential.

5. Adaptation to Environmental Variability:

- Rapid environmental changes due to climate variability make it difficult to rely solely on historical data. CropIQ must continuously adapt to these changes, which requires frequent updates to the model and data sources.

6. Interpretability of Machine Learning Models:

- Deep learning models can act as "black boxes," making it hard to explain why certain crop recommendations are made. This lack of transparency can be a barrier to gaining user trust.

7. Ensuring Regional Relevance:

- Crop characteristics, soil types, and climate conditions vary significantly across regions. The model needs to be tested and fine-tuned for different regions to ensure recommendations are accurate and relevant.

8. Scalability and Maintenance:

- As CropIQ expands to cover larger areas and more users, maintaining system performance and managing data storage needs can be challenging. This requires scalable cloud infrastructure, which can be costly.

9. Promoting User Adoption:

- Convincing farmers to adopt a data-driven approach can be challenging, especially if they are used to traditional methods. Clear communication and user education are needed to build confidence in the system.

By addressing these challenges through targeted strategies—such as improving preprocessing algorithms, simplifying the interface, and educating users on the benefits of data-driven farming—CropIQ can become a more robust and reliable tool for sustainable agriculture.

7.5 Solutions and Improvements

To enhance CropIQ and tackle its challenges, here are solutions and potential improvements

1. Improving Data Quality and Availability:

- Solution: Use data fusion techniques to combine Sentinel-2 with other satellite data sources (e.g., Landsat), which can help fill gaps in coverage due to cloud cover or poor weather.
- Improvement: Implement machine learning algorithms that can perform cloud removal and estimate soil conditions even with partially obscured images.

2. Streamlining Data Processing:

- Solution: Automate preprocessing steps, such as atmospheric correction and noise reduction, using cloud-based processing tools to reduce manual effort and ensure consistency.
- Improvement: Utilize efficient, pre-trained models and reduce processing requirements by only updating data for areas with recent significant environmental changes.

3. Optimizing Model Complexity and Training:

- Solution: Use transfer learning and pre-trained deep learning models, like EfficientNet, which require fewer resources and training data while maintaining high accuracy.
- Improvement: Introduce periodic, targeted retraining focused on regions where the model may be underperforming, rather than retraining the entire model globally.

4. Designing for Limited Internet and Device Access:

- Solution: Develop a lightweight app or offline mode that downloads only essential data for quick access in low-connectivity regions.
- Improvement: Leverage progressive web application (PWA) technology, allowing CropIQ to work offline and sync data when internet access is restored.

5. Adapting to Environmental Variability:

- Solution: Regularly update CropIQ's satellite data and integrate short-term weather forecasts to provide dynamic, real-time recommendations.
- Improvement: Implement predictive analytics to simulate possible environmental changes and adjust recommendations proactively.

6. Enhancing Model Interpretability:

- Solution: Implement explainable AI (XAI) methods to provide insights into the recommendation process, such as showing which soil and environmental factors led to the crop suggestions.
- Improvement: Include a feature in the UI that explains recommendations in simple terms, allowing users to understand and trust the system's guidance.

7. Ensuring Regional Relevance:

- Solution: Calibrate CropIQ with localized data for different regions by working with agricultural extension services and local agronomists to validate recommendations.
- Improvement: Conduct regular validation studies in different regions and adjust the model to ensure it adapts accurately to varying soil types and climates.

8. Scalability and Maintenance:

- Solution: Utilize scalable cloud-based infrastructure to support data storage and processing, making CropIQ easily expandable as usage grows.

- Improvement: Introduce automated monitoring and updates, allowing the system to detect and address issues without manual intervention.

9. Promoting User Adoption:

- Solution: Conduct educational campaigns and training sessions to demonstrate CropIQ's value to farmers and advisors, potentially through government or agricultural organizations.
- Improvement: Create tutorials within the app to guide users through features and provide case studies that showcase CropIQ's effectiveness in improving yield and reducing costs.
- By implementing these solutions, CropIQ can evolve into a robust, scalable, and user-friendly tool that effectively supports sustainable and profitable agriculture. These improvements will also help build trust and ensure the system meets diverse user needs across varying regions and conditions.

CHAPTER -VIII

CONCLUSION

The "Crop Recommendation System Using Satellite Images and Deep Learning" effectively combines satellite technology and machine learning to improve farming decisions. By analyzing Sentinel-2 imagery and using deep learning models like EfficientNet, the system provides accurate crop recommendations based on soil type and environmental conditions. This data-driven approach enables farmers to optimize yields and sustainability by making informed decisions on crop selection, which boosts productivity and profitability.

Despite its promising results, the system has limitations, such as a reliance on accurate satellite data and the lack of integration with real-time factors like weather conditions or market demands. Nevertheless, the system provides a solid foundation for further advancements in precision agriculture by addressing the challenges of traditional, manual crop selection methods. The technology's potential to streamline farming practices makes it an important tool in modern agriculture.

FUTURE SCOPE

Several enhancements could make the system more effective:

Real-time Weather Data Integration: Incorporating real-time weather forecasts would help refine crop recommendations based on predicted climatic conditions.

Market Demand Consideration: Future versions could factor in real-time market data to suggest crops that offer higher profitability.

Pest and Disease Monitoring: Integrating pest and disease prediction could help farmers make better decisions about pest-resistant crops or necessary interventions.

Mobile Application: Developing a mobile app would increase accessibility, allowing farmers to easily input location data and receive crop recommendations.

Global Expansion: The system could be adapted to other agricultural regions worldwide, providing tailored recommendations for different climates and soil

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PROJECT AUDIT REPORT

This is to certify that the project work entitled “CropIQ: Enhancing Crop Selection with Satellite and Drone Image Analysis” categorized as an internal project done by P. Durga Prasadu (9921004520), P. Anjana Sree(9921004816), P. Manideep(9921004564), P.Lohitha Lakshmi(9921004563) of the Department of Computer Science and Engineering, under the guidance of Dr. Anitha Ponraj during the Even semester of the academic year 2023 - 2024 are as per the quality guidelines specified by IQAC.

Quality Grade

Deputy Dean (IQAC)

Administrative Quality Assurance

Dean (IQAC)